

IMO(21st) – 1979

First Day

1. Let p, q be natural numbers such that $\frac{p}{q} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{1318} + \frac{1}{1319}$. Prove that p is divisible by 1979.
2. A prism with pentagons $A_1A_2A_3A_4A_5$ and $B_1B_2B_3B_4B_5$ as top and bottom faces is given. Each side of the two pentagons and each of the line segments A_iB_j for all $i, j = 1, 2, \dots, 5$ is colored either red or green. Every triangle whose vertices are vertices of the prism and whose sides have all been colored has two sides of a different color. Show that all 10 sides of the top and bottom faces are the same color.
3. Two circles in a plane intersect. Let A be one of the points of intersection. Starting simultaneously from A two points move with constant speeds, each point travelling along its own circle in the same sense. The two points return to A simultaneously after one revolution. Prove that there is a fixed point P in the plane such that, at any time, the distances from P to the moving points are equal.

Second Day

4. Given a plane π , a point P in this plane and a point Q not in π , find all points R in π such that the ratio $(QP+PA)/QR$ is maximum.
5. Find all real numbers a for which there exist non-negative real numbers x_1, x_2, x_3, x_4, x_5 satisfying the relations $\sum_{k=1}^5 kx_k = a, \sum_{k=1}^5 k^3 x_k = a^2, \sum_{k=1}^5 k^5 x_k = a^3$.
6. Let A and E be opposite vertices of a regular octagon. A frog starts jumping at vertex A . From any vertex of the octagon except E , it may jump to either of the two adjacent vertices. When it reaches vertex E , the frog stops and stays there. Let a_n be the number of distinct paths of exactly n jumps ending at E . Prove that $a_{2n-1} = 0, a_{2n} = \frac{1}{\sqrt{2}}(x^{n-1} - y^{n-1}), n = 1, 2, 3, \dots$ where $x = 2 + \sqrt{2}$ and $y = 2 - \sqrt{2}$.
Note that: A path of n jumps is a sequence of vertices $(P_0, P_1, \dots, P_n)$ such that:
(a) $P_0 = A, P_n = E$;
(b) For every $i, 0 \leq i \leq n-1, P_i$ is distinct from E ;
(c) For every $i, 0 \leq i \leq n-1, P_i$ and P_{i+1} are adjacent.

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